

Gain > 80 dB

Settling time = 2 ns (?)

u should be able to hit 70dB gain easily using folded cascode design.... 46dB sound like u dont have transistors biased correctly. use 5-10 times minimum length for current sources/rail devices and 2-3 times minimum L for your cascodes. this will give u good output impedance.

NMOS

$$\frac{g_m}{I_D} = 16$$

$$\frac{g_m}{I_D} = 10$$

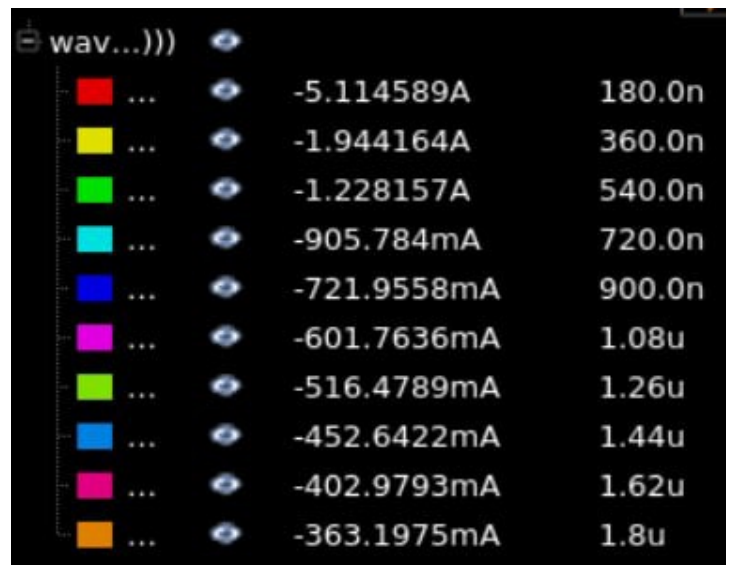
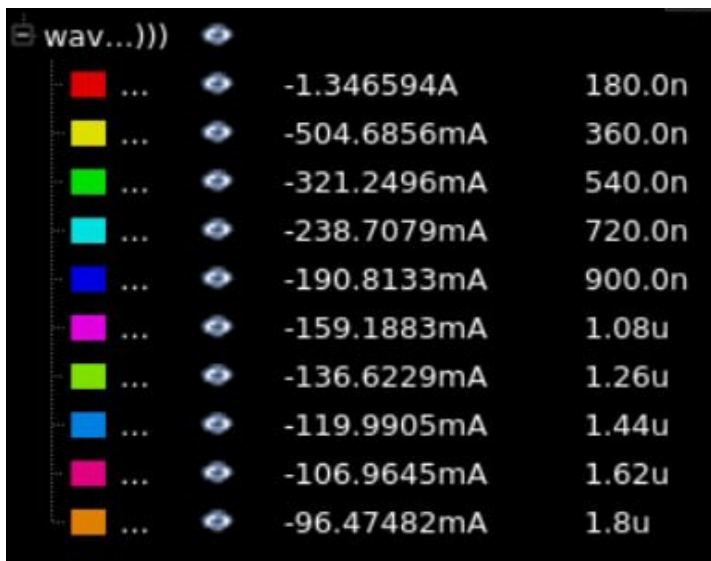
waveVsWave...AR("Wn"))	
waveVsWa...("Wn"))	564.4034mA 1.8u
waveVsWa...("Wn"))	628.6574mA 1.62u
waveVsWa...("Wn"))	706.9788mA 1.44u
waveVsWa...("Wn"))	811.6382mA 1.26u
waveVsWa...("Wn"))	945.7941mA 1.08u
waveVsWa...("Wn"))	1.135058A 900.0n
waveVsWa...("Wn"))	1.414752A 720.0n
waveVsWa...("Wn"))	1.887888A 540.0n
waveVsWa...("Wn"))	2.883991A 360.0n
waveVsWa...("Wn"))	6.478222A 180.0n

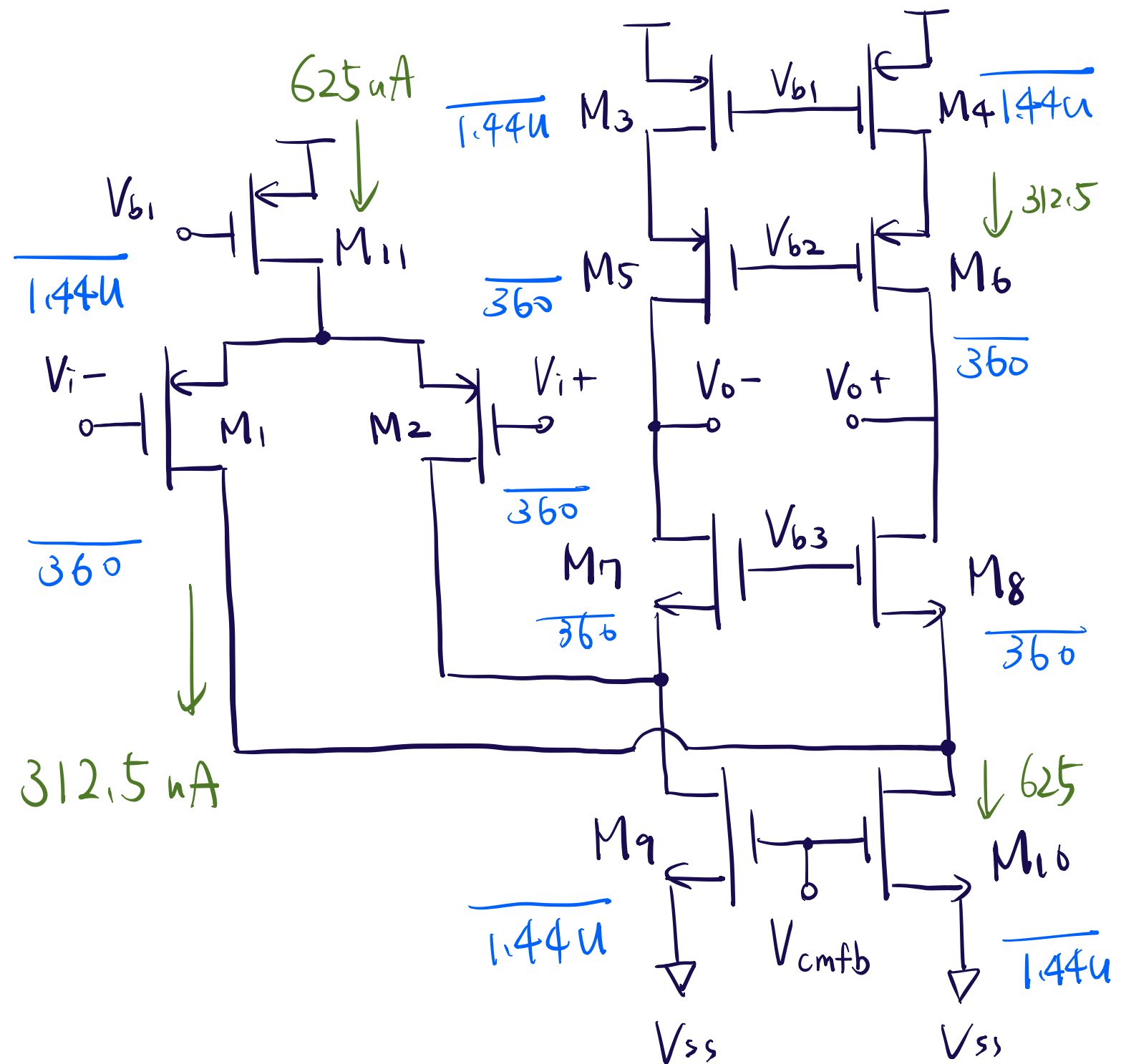
waveVsWave...AR("Wn"))	
waveVsWa...("Wn"))	1.943201A 1.8u
waveVsWa...("Wn"))	2.163649A 1.62u
waveVsWa...("Wn"))	2.43704A 1.44u
waveVsWa...("Wn"))	2.789339A 1.26u
waveVsWa...("Wn"))	3.2629A 1.08u
waveVsWa...("Wn"))	3.924556A 900.0n
waveVsWa...("Wn"))	4.924768A 720.0n
waveVsWa...("Wn"))	6.632693A 540.0n
waveVsWa...("Wn"))	10.19208A 360.0n
waveVsWa...("Wn"))	21.4431A 180.0n

PMOS

$$\frac{g_m}{I_D} = 16$$

$$\frac{g_m}{I_D} = 10$$





$$\frac{W_3}{I_{bias}} = \frac{W_{12}}{200}$$

Assume $C_L = 4 \text{ pF}$

① $GBW = 200 \text{ MHz} \Rightarrow g_{m1} = 5 \text{ mA/V}$

$$\frac{g_{m1}}{I_{D1}} = 16 \cdot \frac{I_{D1}}{W_1} = 0.5 \quad \left(\frac{W}{L}\right)_{1,2} = \frac{312\mu}{360\text{n}}$$

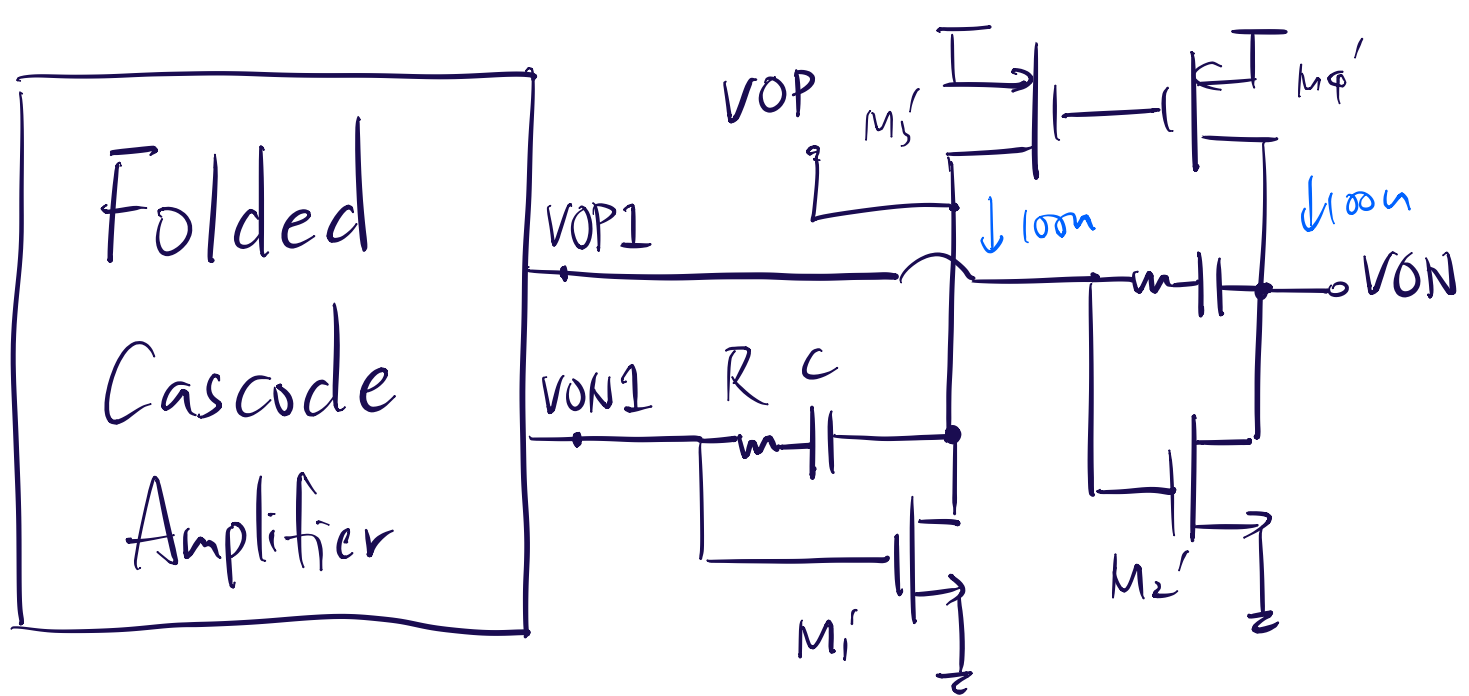
$$\frac{g_{m11}}{I_{D11}} = 10 \cdot \frac{I_{D11}}{W_{11}} = 0.45 \quad \left(\frac{W}{L}\right)_{11} = \frac{1390\mu}{1.44\mu}$$

$$\frac{g_{m3}}{I_{D3}} = 10 \cdot \frac{I_{D3}}{W_3} = 0.45 \quad \left(\frac{W}{L}\right)_3 = \frac{695\mu}{1.44\mu}$$

$$\frac{g_{m5}}{I_{D5}} = 10 \cdot \frac{I_{D5}}{W_5} = 1.94 \quad \left(\frac{W}{L}\right)_5 = \frac{161\mu}{360\text{n}}$$

$$\frac{g_{m7}}{I_{D7}} = 10 \cdot \frac{I_{D7}}{W_7} = 10.2 \quad \left(\frac{W}{L}\right)_7 = \frac{30\mu}{360\text{n}}$$

$$\frac{g_{m9}}{I_{D9}} = 10 \cdot \frac{I_{D9}}{W_9} = 2.44 \quad \left(\frac{W}{L}\right)_9 = \frac{256\mu}{1.44\mu}$$

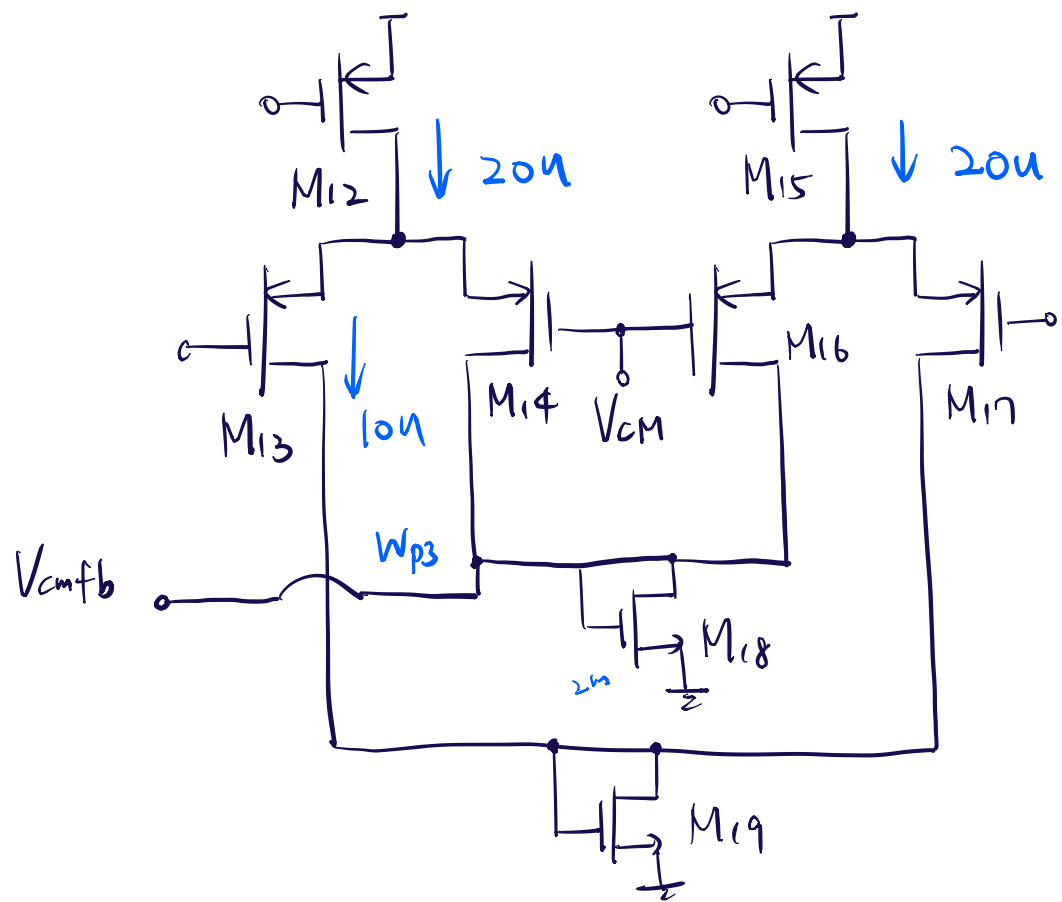


$$\frac{g_{m1'}}{I_{b1'}} = 10$$

$$\frac{I_{b1'}}{W_{1'}} = 10.2 \quad \left(\frac{W}{L}\right)'_1 = \frac{10 \mu}{360 \mu}$$

$$\frac{g_{m3'}}{I_{b3'}} = 10 \quad \frac{I_{b3'}}{W_{3'}} = 1.94$$

$$\left(\frac{W}{L}\right)'_3 = \frac{52 \mu}{360 \mu}$$



$$\frac{g_{m12}}{I_{D12}} = 10, \quad \frac{I_{D12}}{W_{12}} = 0.45, \quad \left(\frac{W}{L}\right)_{12} = \frac{44\mu}{1.44\mu}$$

$$\frac{g_{m13}}{I_{D13}} = 16, \quad \frac{I_{D13}}{W_{13}} = 0.5, \quad \left(\frac{W}{L}\right)_{13} = \frac{20\mu}{360\mu}$$

$\tau = RC$, settling time

GBW {

400 μ $\rightarrow$ 44 MV/s

5.45 mA $\rightarrow$ 600 MV/s